MINDSHIELD

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ABSTRACT

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MindShield is an interactive web-based cybersecurity awareness platform designed to educate users about social engineering threats. The project addresses the growing impact of phishing, impersonation, and online scams by providing structured learning modules, scenario-based simulations, and quizzes with immediate feedback. The system integrates user authentication, progress tracking, and content management features to support different user roles including learners and administrators. From a technical perspective, MindShield employs a modern web architecture supported by cloud services for secure data storage, authentication, and scalable delivery. The expected outcome is to improve users' ability to recognize and respond to social engineering attempts, thereby reducing human-related security risks.

ACKNOWLEDGEMENTS

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I would like to express my sincere gratitude to **Nabeel Al-Assaf** for the guidance, support, and valuable feedback provided throughout this project. I also thank the Department of CS (COMPUTER SCIENCE)and the King Abdullah II School of Information Technology for the learning environment and resources. Finally, I am grateful to my family and friends for their continuous encouragement and support.

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LIST OF ABBREVIATIONS

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Abbreviation	Meaning
API	Application Programming Interface
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
NFR	Non-Functional Requirement
FR	Functional Requirement
UI	User Interface
UML	Unified Modeling Language

1.0 INTRODUCTION

1.1 Project Motivation

From the early days of human interaction to the modern digital era, manipulation through deception has always existed. Today, with the rapid expansion of the internet, social media, and digital services, social engineering attacks have become among the most common and dangerous cybersecurity threats. These attacks target human behavior rather than technical vulnerabilities, leading to identity theft, financial loss, data breaches, and significant stress for individuals and organizations.

Despite advancements in technical security solutions, many users remain vulnerable due to limited awareness, insufficient training, and an incomplete understanding of how social engineering techniques operate. Phishing emails, fake websites, impersonation attempts, and malicious social media interactions continue to increase because human awareness is often the weakest link in the security chain.

MindShield is designed to address these challenges by providing a centralized, user-focused awareness platform that educates users about social engineering threats, enhances their ability to recognize and respond to attacks, and promotes safer digital behavior. By combining structured learning, practice, and feedback, MindShield aims to reduce human-related security risks and strengthen the overall cybersecurity posture of its users.

1.2 Problem Statement

- **Inconsistent learning sources:** Users often rely on multiple unconnected resources (social media posts, videos, informal advice) to understand social engineering threats, leading to confusion and inconsistent security practices.
- **Forgetting safe practices over time:** Without continuous reinforcement, users may forget essential behaviors such as verifying links, identifying phishing indicators, and applying basic cyber hygiene.
- **Lack of structured learning and assessment:** Many awareness resources are not organized into progressive modules, and they lack measurable assessments and feedback that confirm user understanding.
- **Limited visibility for program owners:** In organizational settings, administrators may lack an easy way to monitor participation, measure performance, and identify common weaknesses across users.

1.3 Project Aim and Objectives

Aim: Develop a web-based awareness and education platform that helps users recognize and respond to social engineering threats through structured learning modules, quizzes, feedback, and progress tracking, with administrator capabilities for content management and analytics.

Objectives:

1. Implement secure user registration and authentication.
2. Provide structured learning modules that cover common social engineering attacks (e.g., phishing, impersonation, scams) with clear navigation.
3. Deliver scenario-based quizzes with instant feedback and the ability to retake quizzes for reinforcement.
4. Track and visualize user learning progress across modules and assessments.
5. Generate a completion status or certificate for users who successfully finish required learning content.

6. Provide administrator functions to manage content and questions and to view reports/analytics on user participation and performance.
7. Support usability features such as search and notifications to improve accessibility and engagement.

1.4 Project Scope

MindShield focuses on a learning-first web platform for social engineering awareness. The scope of the current version includes:

- **User registration and login** with secure authentication.
- **Learning modules** organized by topic with smooth navigation and content access.
- **Scenario-based quizzes** to assess understanding, with **instant feedback** and **retake** support.
- **Progress tracking** to display module completion and quiz outcomes.
- **User profile management** (basic profile viewing/updating).
- **Completion status / certificate** upon satisfying completion conditions.
- **Role-based access control** separating learner and administrator capabilities.
- **Admin content management** (modules, lessons, and questions) and **reporting/analytics**.
- **Notifications** related to new content, incomplete learning items, or results.
- **Search functionality** to locate topics and modules quickly.

Out of scope for the current version (future enhancements): multi-language localization, offline access to learning content, and community discussion forums.

1.5 Software and Hardware Requirements

Table 1.1: Software Requirements

Component	Requirement
Front-End Development	HTML, CSS, and JavaScript (interactive UI, quizzes, and client-side behavior).
Backend Services	Firebase (Authentication, Firestore, Cloud Messaging, Storage) and Firebase CLI (deployment and local emulators).
Code Editor	Visual Studio Code (recommended extensions for web development).
Version Control	GitHub.
Project Management	Jira.

Table 1.2: Hardware Requirements

Component	Requirement
Development Computers	1–4 computers for the team (Windows/Linux acceptable), minimum 8 GB RAM each.
Storage	At least 500 GB free disk space (tools, dependencies, build artifacts, and backups).
Network	Stable broadband Internet connection (Firebase services and real-time updates).

1.6 Project Limitations

- **Internet dependency:** MindShield requires an active internet connection for real-time updates and cloud-backed features (authentication, database, notifications), which may limit usability in areas with unstable connectivity.
- **Budget constraints:** Development and deployment costs (including cloud usage and cross-device testing) may restrict rapid expansion and advanced feature additions.

- **Device/browser compatibility:** Older devices and outdated browsers may not fully support modern web application features, potentially limiting accessibility for some users.

1.7 Project Expected Output

- A functional web-based application for social engineering awareness (MindShield).
- Structured learning modules and scenario-based quizzes with detailed feedback.
- Progress tracking, completion status, and certificate generation.
- Administrator dashboard for content management and performance analytics.
- User documentation that explains platform usage and recommended cyber hygiene practices.

1.8 Project Schedule

The project schedule is summarized in Table 1.3. A detailed Gantt chart can be inserted here as a figure (e.g., *Figure 1.1*) in the final report.

Table 1.3: High-Level Project Schedule (Template)

Phase	Planned Period (Fill/Adjust)
Requirements Engineering (Ch. 1–3 drafting)	Weeks 1–4
System Design (Ch. 4)	Weeks 5–7
Implementation (Ch. 5)	Weeks 8–11
Testing and Installation (Ch. 6)	Weeks 12–13
Final Documentation and Packaging	Week 14

1.9 Report Outline

This report is organized into chapters as follows:

1. **Introduction** — presents motivation, problem statement, aims and objectives, scope, requirements, limitations, expected outputs, schedule, and outline.
2. **Related Existing Systems** — reviews existing systems/products, identifies gaps, and summarizes proposed improvements addressed by MindShield.

3. **System Requirements Engineering and Analysis** — presents feasibility, requirements gathering, targeted users, and functional/non-functional requirements.
4. **System Design** — presents context/DFD/ERD, UML diagrams, and database design.
5. **System Implementation** — describes implementation details and tools used.
6. **System Testing and Installation** — presents testing approach, installation steps, and user manual.
7. **Conclusions and Future Work** — concludes the project and proposes future enhancements.

2.0 RELATED EXISTING SYSTEMS

2.1 Overview of Existing Products

Reviewing related existing systems helps identify what capabilities already exist, what limitations remain, and what features are most valuable for the intended users. This analysis supports defining clearer and more realistic requirements for MindShield, a learning-first web platform focused on social engineering awareness through modules, quizzes, feedback, and progress analytics.

2.1.1 KnowBe4 Security Awareness Training (KSAT)

KnowBe4 is a widely used security awareness training platform that provides a large content library and simulated phishing campaigns with reporting for organizations.

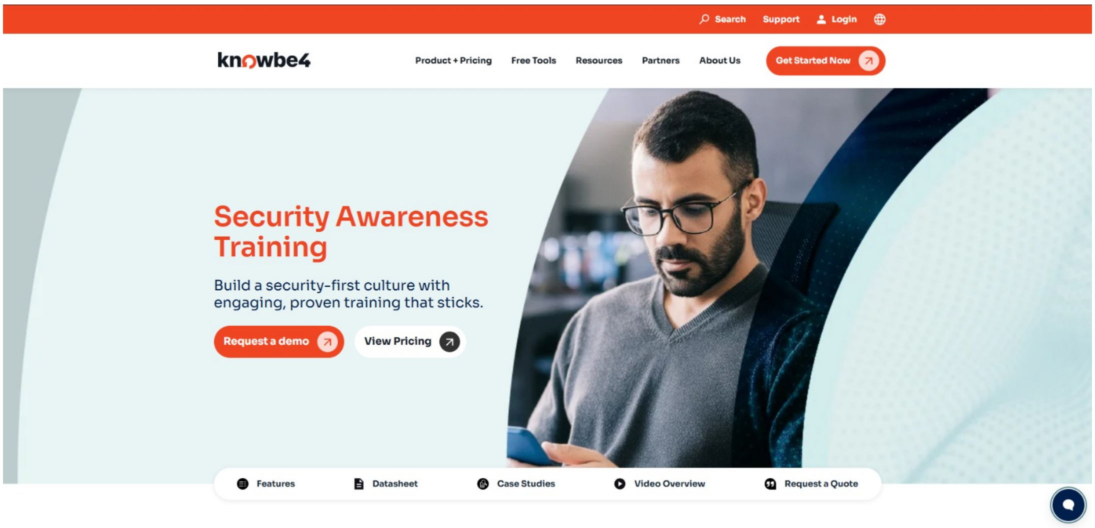

Figure 2.1: KnowBe4 Security Awareness Training (KSAT)

Strength points:

- Large security awareness content library and simulated phishing capability.
- Supports multiple languages and localized content.
- Includes dashboards and reporting to track user behavior and training outcomes.

Weaknesses:

- Primarily enterprise-focused and subscription-based, which can be costly for small organizations or student projects.
- Customization for local or university contexts may require additional setup and administration.
- The platform can feel heavy for users seeking quick, beginner-friendly learning in a lightweight web format.

2.1.2 Proofpoint Security Awareness Training

Proofpoint provides training and phishing simulations with a focus on measuring and reducing human-related security risk through assessments and analytics.

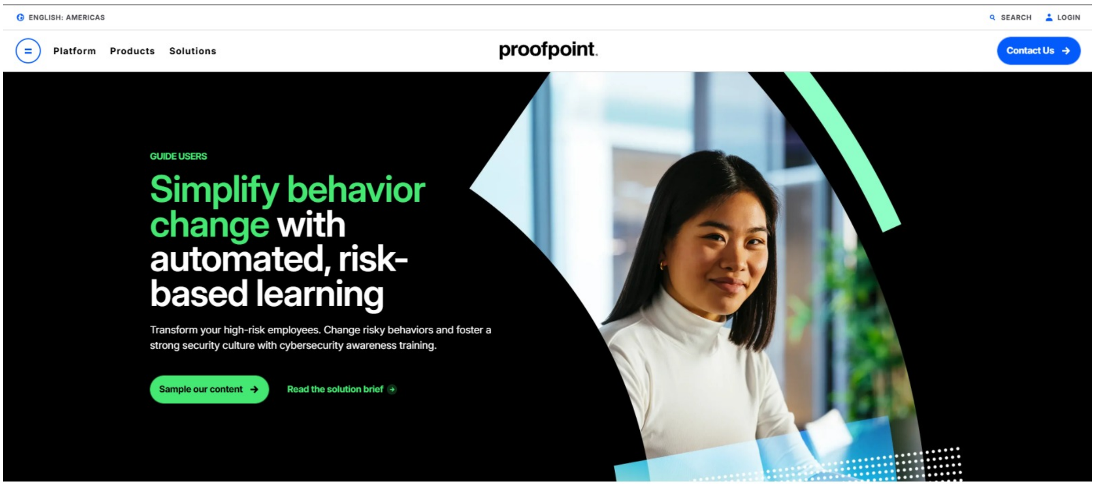

Figure 2.2: Proofpoint Security Awareness Training

Strength points:

- Simulations and assessments to evaluate awareness and susceptibility.
- Risk-based insights and dashboards to highlight high-risk behaviors.
- Reporting features to support awareness program management.

Weaknesses:

- Designed for organizations with structured programs and budgets.
- Less suitable for lightweight public-facing awareness for students and general users.

- More compliance-oriented than learning-first, which may reduce engagement for beginners.

2.2 Problems and Weaknesses (Common Gaps)

While existing solutions provide valuable training and simulation capabilities, several common limitations appear when compared to the needs and scope of MindShield:

1. **High cost / enterprise orientation:** Many platforms are subscription-based and designed for organizations with structured training cycles.
2. **Limited beginner-friendly learning focus:** Some systems emphasize testing and compliance rather than progressive learning (lesson → practice → quiz → feedback).
3. **Limited accessibility for general users:** Many solutions are internal corporate tools rather than open awareness websites.
4. **Customization vs. simplicity trade-off:** Powerful tools can be complex and reduce usability for beginner audiences.

2.3 Proposed Solutions in MindShield

MindShield addresses these gaps by providing a focused, interactive, and beginner-friendly web platform dedicated to social engineering awareness:

1. **Short, topic-based modules** for common social engineering techniques.
2. **Scenario-based quizzes with instant feedback** to reinforce correct decisions.
3. **Progress tracking and completion status** to monitor learning progression.
4. **Certificate generation** to improve motivation and engagement.
5. **Admin content management and analytics** to control modules/quizzes and monitor usage trends.
6. **Search and notifications** to improve content accessibility and user engagement.

2.4 Summary

Enterprise awareness platforms provide strong capabilities, but they often come with cost and complexity. MindShield focuses on a simpler learning-first approach combining modules, quizzes, feedback, progress tracking, and administrator analytics in one interactive website tailored to social engineering awareness goals.

3.0 SYSTEM REQUIREMENTS ENGINEERING AND ANALYSIS

3.1 Feasibility Study

A feasibility study was conducted to determine the viability of developing MindShield. The study examined technical, operational, and economic aspects of the project.

Technical feasibility: assesses whether the project can be implemented with existing or readily available technology. MindShield follows a standard interactive web-application architecture, including front-end development (HTML, CSS, JavaScript) for user interaction and quizzes, and back-end services (Firebase Authentication and Firestore) to manage users, content, progress tracking, and quiz results. These technologies are mature, well-documented, and suitable for the project requirements. Based on the development team's skills and the availability of tools, the project is technically feasible.

Operational feasibility: evaluates whether the proposed system will be useful, usable, and compatible with the target users' environments. The primary goal is to educate users, and an interactive website provides high accessibility across common devices and browsers. MindShield will prioritize intuitive navigation and clear instructional design to ensure adoption by beginner audiences. Therefore, the project is operationally feasible.

Economic feasibility: determines whether benefits outweigh costs. Costs include development time, cloud service usage, and ongoing maintenance. Benefits include improved awareness and reduced likelihood of successful social engineering incidents. A high-level cost-benefit view indicates the project is economically feasible, as cloud costs are scalable and the expected value of awareness improvement is significant.

3.2 Targeted Users

MindShield targets two primary user roles that align with the system design (use-case actors):

- **Learners (End Users):** Individuals who use MindShield to learn about social engineering threats, complete modules and quizzes, receive feedback, and track progress. This group includes general internet users, students, educators, and employees seeking awareness training.
- **Administrators (Content Managers):** Users responsible for managing educational content and quiz questions, monitoring user participation and performance analytics, and ensuring the platform remains up-to-date.

3.3 Functional Requirements Definition and Specification

Functional requirements define the actions and services that MindShield must provide to learners and administrators.

Table 3.1: Functional Requirements

Requirement ID	Requirement Name	Description
FR-001	User Registration and Authentication	The system must allow users to register with a unique email and password and securely log in.
FR-002	Content Access and Navigation	The system must allow learners to browse educational modules structured around different social engineering threats.
FR-003	Interactive Quiz Module	The system must present scenario-based quizzes to test user understanding and decision-making skills.
FR-004	Progress Tracking	The system must track and display the learner's progress through modules and quiz completion status.
FR-005	Feedback Mechanism	Upon quiz completion, the system must provide immediate, detailed feedback explaining correct and incorrect responses.
FR-006	Content Management	The system must allow administrators to add, edit, and categorize educational content and quiz questions.
FR-007	User Profile Management	The system must allow learners to view and update their personal profile information (e.g., name and preferences).
FR-008	Module Completion Certification	The system must generate a completion status or digital certificate for learners who successfully finish required modules and quizzes.
FR-009	Quiz Retake Functionality	The system must allow learners to retake quizzes to improve their scores and reinforce learning.
FR-010	Reporting and Analytics	The system must provide administrators with reports and analytics on participation, quiz performance, and overall progress trends.
FR-011	Notification System	The system must notify learners about newly added content, incomplete modules, or quiz results via on-site notifications or email (as applicable).
FR-012	Search Functionality	The system must allow learners to search for specific topics, modules, or keywords related to social engineering threats.
FR-013	Role-Based Access Control	The system must differentiate between learner and administrator roles and restrict access to administrative features accordingly.

3.4 Non-Functional Requirements

Non-functional requirements describe the quality attributes and constraints that affect system performance, security, usability, and reliability.

Table 3.2: Non-Functional Requirements

Requirement ID	Category	Description
NFR-001	Performance	The system must load learning content and quiz pages within 3 seconds under normal and peak usage conditions.
NFR-002	Security	The system must protect user data in transit and at rest and mitigate common web threats (e.g., injection and broken access control).
NFR-003	Reliability	The system must ensure an uptime of at least 99.5% and save user progress and quiz results without data loss.
NFR-004	Usability	The system must provide an intuitive interface that allows new users to start learning without external assistance.
NFR-005	Scalability	The system must support growth in concurrent users without noticeable performance degradation.
NFR-006	Maintainability	The system must be modular and documented to allow efficient maintenance and content expansion.
NFR-007	Compatibility	The system must function correctly across modern browsers.
NFR-008	Availability	The system must be accessible 24/7, excluding scheduled maintenance communicated in advance.
NFR-009	Data Integrity	The system must ensure accuracy and consistency of user data during storage, retrieval, and updates.

3.5 Usability and User Experience Goals

Usability and user experience (UX) goals are critical for an educational platform to ensure engagement and effective learning:

- **Learnability:** New users must be able to navigate the website and start the first module without requiring external help.
- **Efficiency:** Returning users should quickly access specific modules or past results with minimal steps (e.g., within three clicks from the homepage).
- **Satisfaction:** The interface must be visually clear and supportive of learning, and it should

encourage continued engagement.

- **Error Prevention:** The system must use clear labels and confirmations to prevent accidental loss of quiz progress.

4.0 SYSTEM DESIGN

This chapter presents the key aspects of the system design, including the context diagram, data flow diagram, entity relationship diagram, UML use case diagrams, UML sequence diagrams, UML class diagram, and database design for MindShield. The sequence diagrams in this chapter correspond to the functional requirements listed in Table 3.1.

4.1 System Context Diagram

Figure 4.1 presents the system context diagram, showing the system boundary and the main external entities interacting with MindShield.

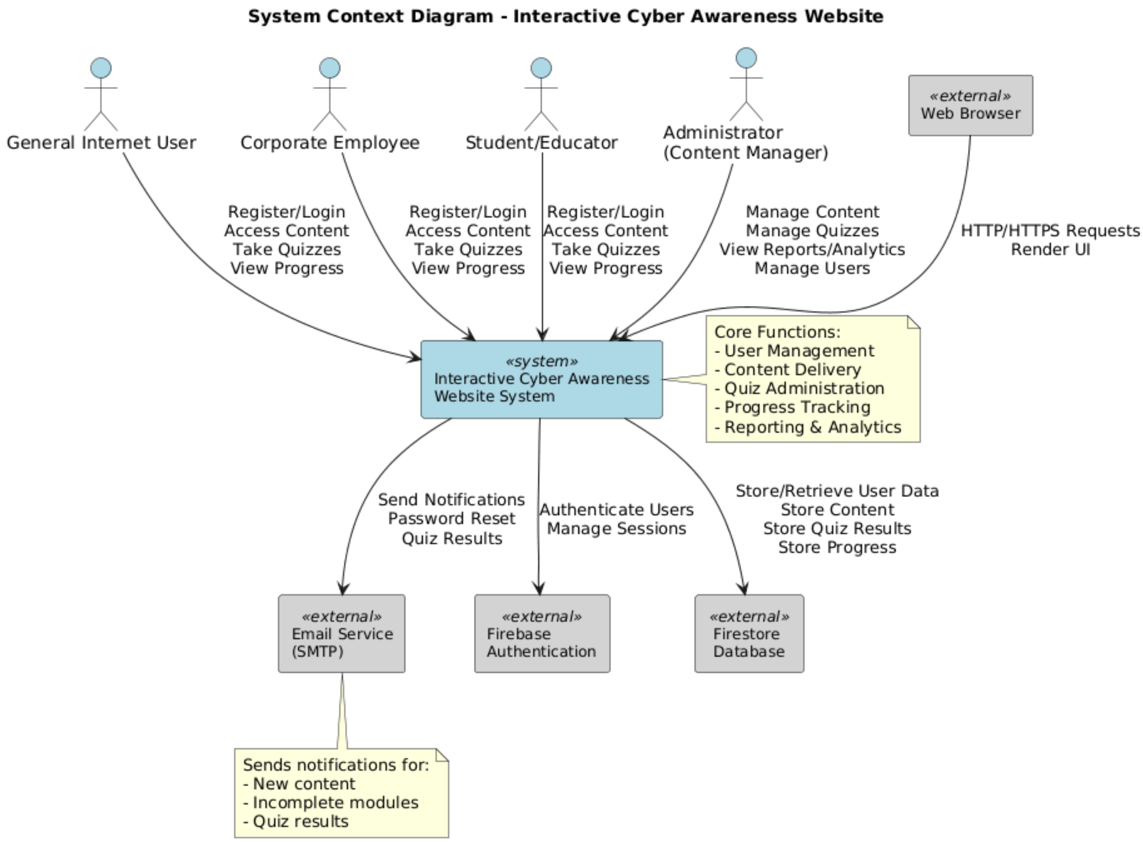

Figure 4.1: System Context Diagram

4.2 Context Diagram

Figure 4.2 illustrates the context diagram of the proposed system and the primary interactions between the users and MindShield.

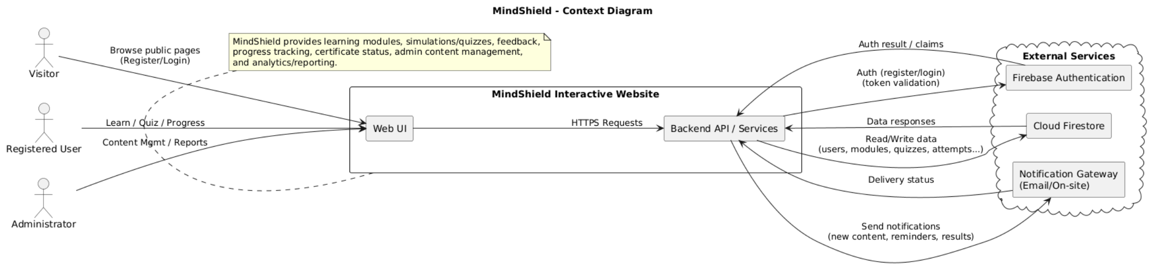

Figure 4.2: Context Diagram

4.3 Data Flow Diagram

This section presents the data flow diagram (DFD) describing how data moves between system processes and data stores.

4.3.1 DFD Level 0

Figure 4.3 shows the DFD Level 0 of the system.

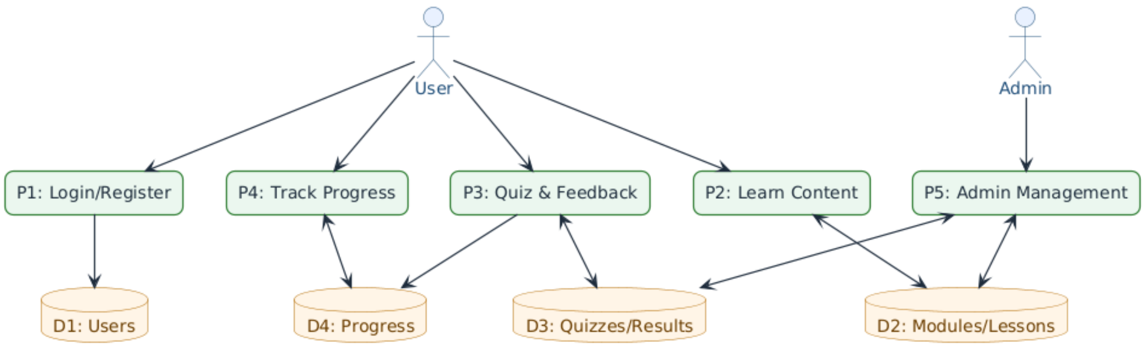

Figure 4.3: DFD Level 0

4.4 Entity Relationship Diagram

Figure 4.4 presents the entity relationship diagram (ERD), describing the main entities and their relationships in the system.

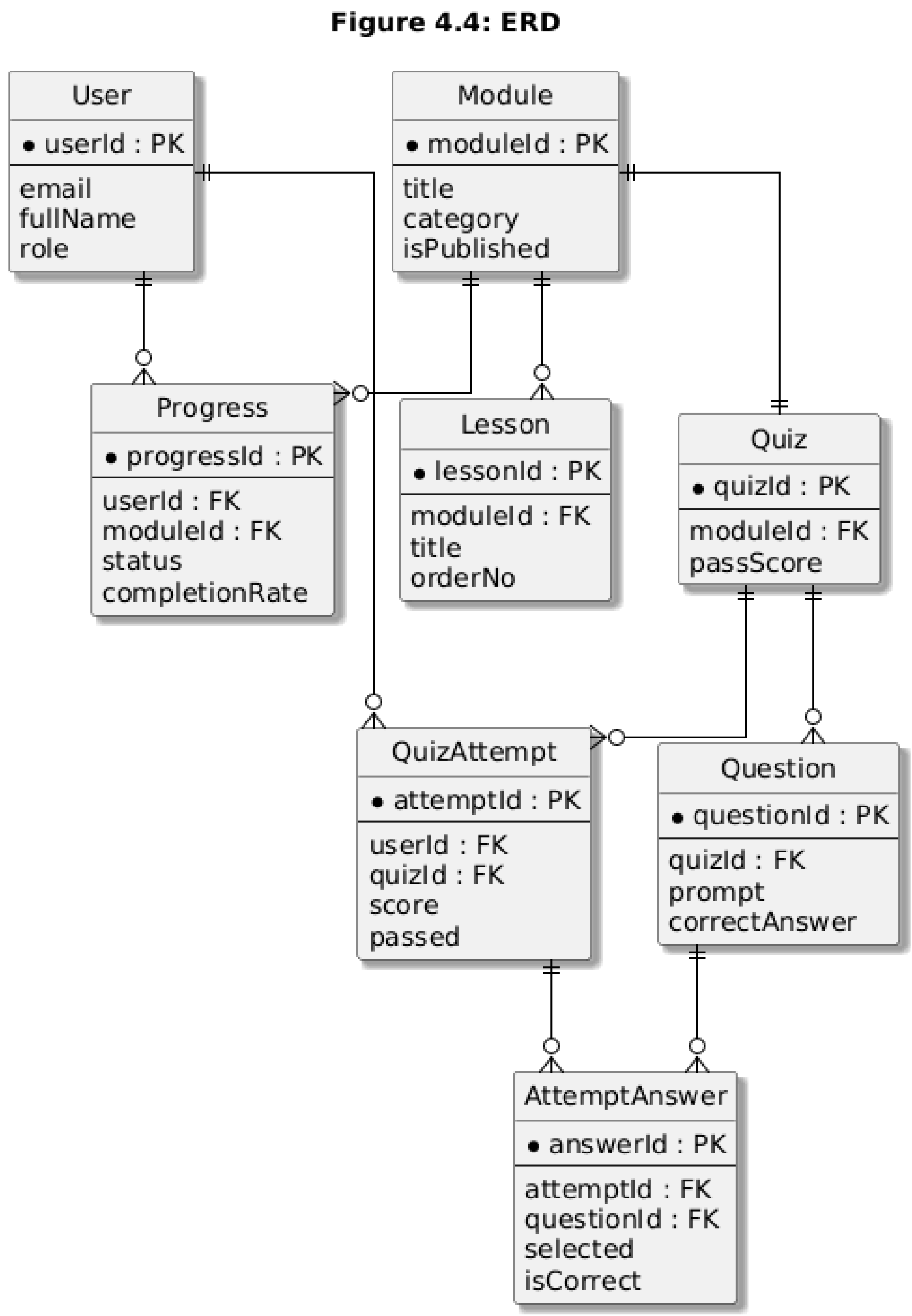

Figure 4.4: Entity Relationship Diagram (ERD)

4.5 UML Use Case Diagrams

This section presents the system use cases from the perspective of each primary actor (Administrator and Learner).

4.5.1 Use Case Diagram (Admin)

Figure 4.5 presents the UML use case diagram for the administrator.

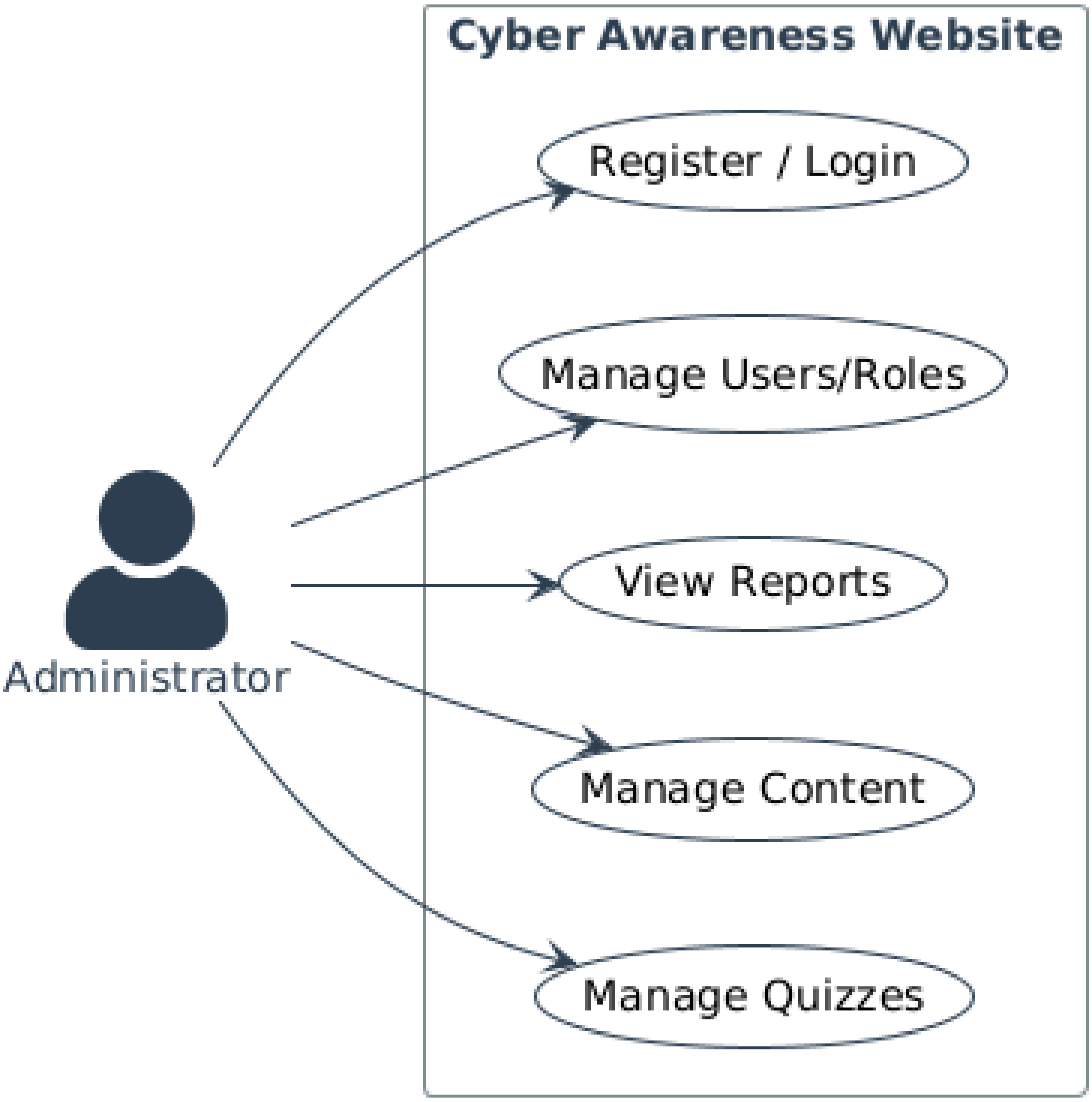

Figure 4.5: Use Case Diagram (Admin)

4.5.2 Use Case Diagram (Learner)

Figure 4.6 presents the UML use case diagram for the learner.

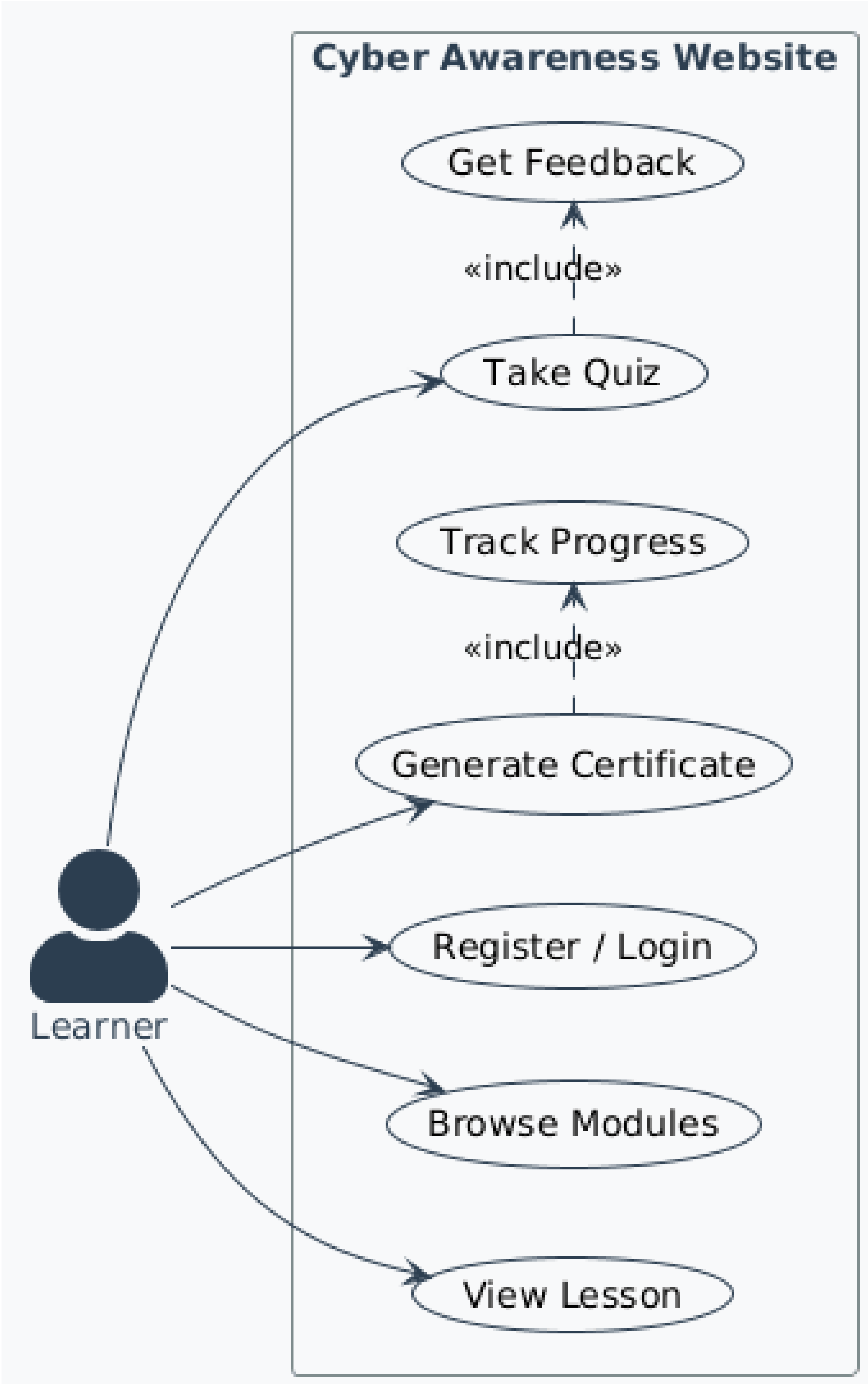

Figure 4.6: Use Case Diagram (Learner)

4.6 UML Sequence Diagrams

This section presents a dedicated UML sequence diagram for each functional requirement (FR), illustrating the interaction flow between the actor, the system components, and the data services.

4.6.1 UML Sequence Diagram (FR-001)

Figure 4.7 presents the UML sequence diagram for FR-001.

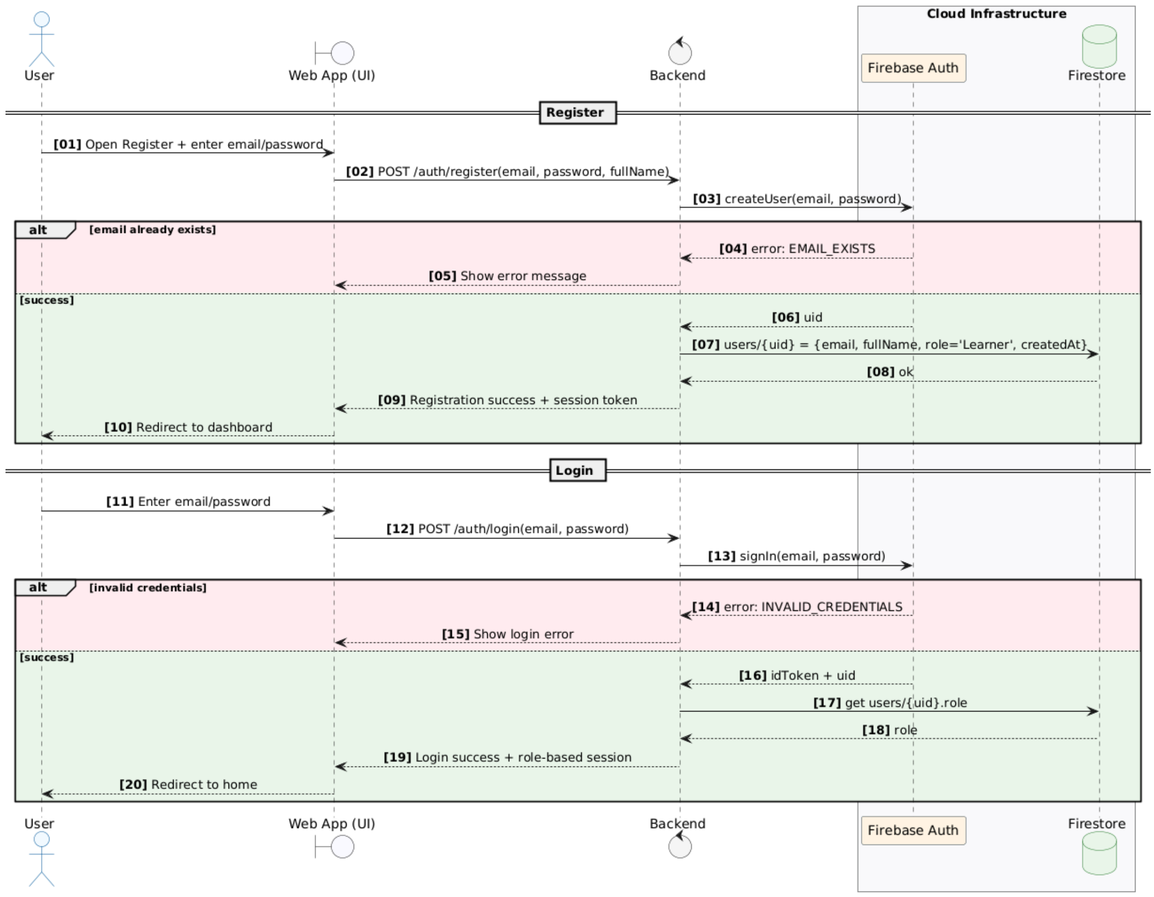

Figure 4.7: UML Sequence Diagram (FR-001)

4.6.2 UML Sequence Diagram (FR-002)

Figure 4.8 presents the UML sequence diagram for FR-002.

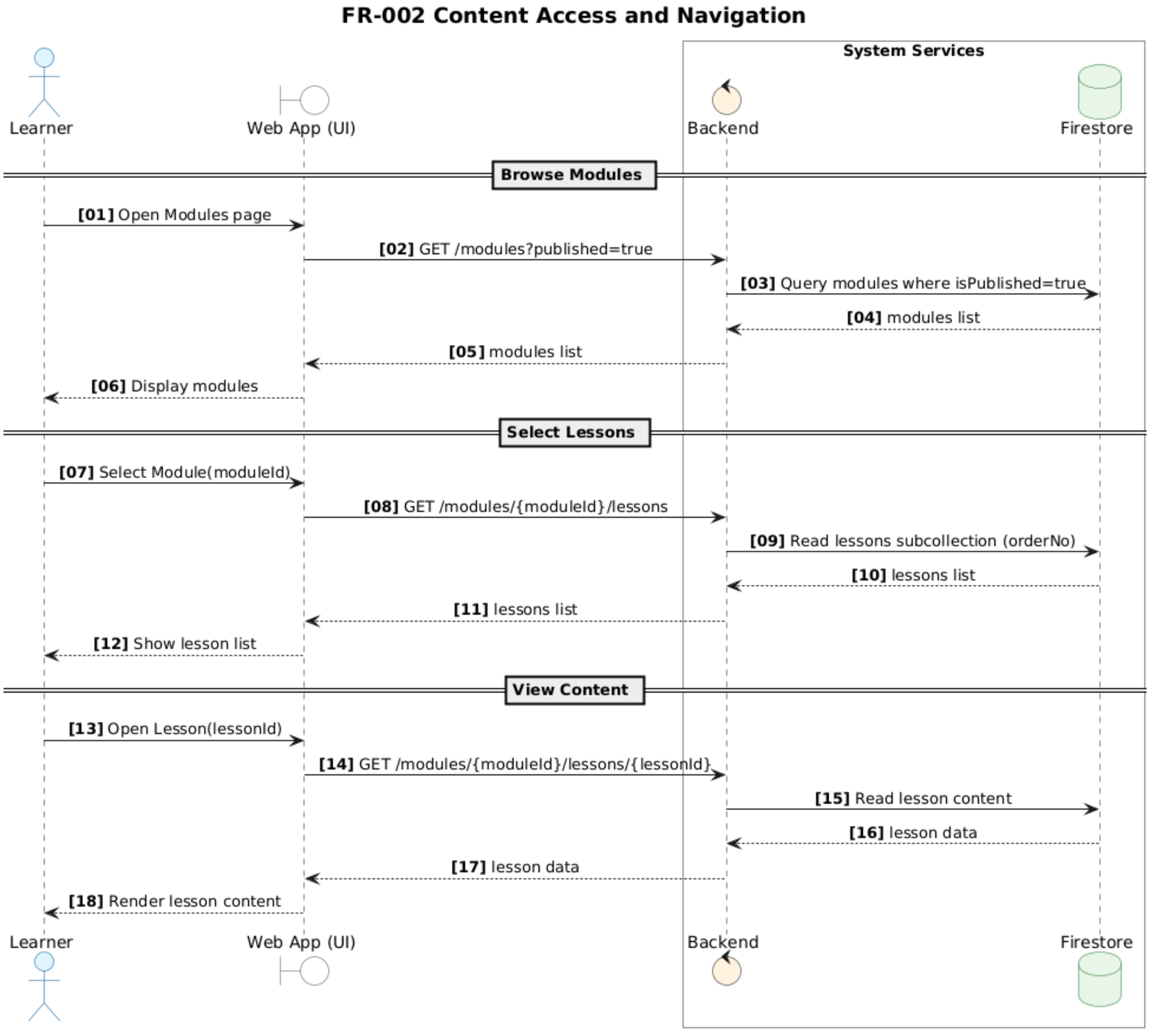

Figure 4.8: UML Sequence Diagram (FR-002)

4.6.3 UML Sequence Diagram (FR-003)

Figure 4.9 presents the UML sequence diagram for FR-003.

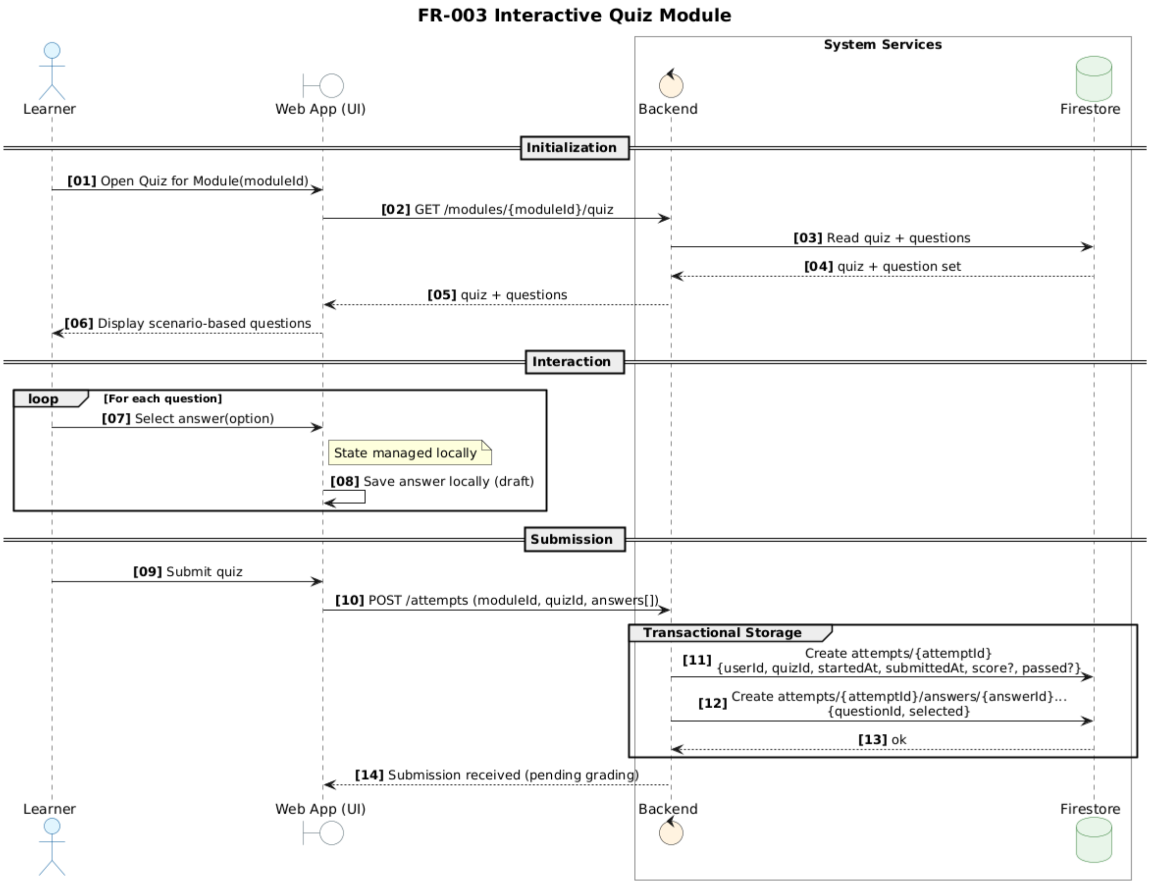

Figure 4.9: UML Sequence Diagram (FR-003)

4.6.4 UML Sequence Diagram (FR-004)

Figure 4.10 presents the UML sequence diagram for FR-004.

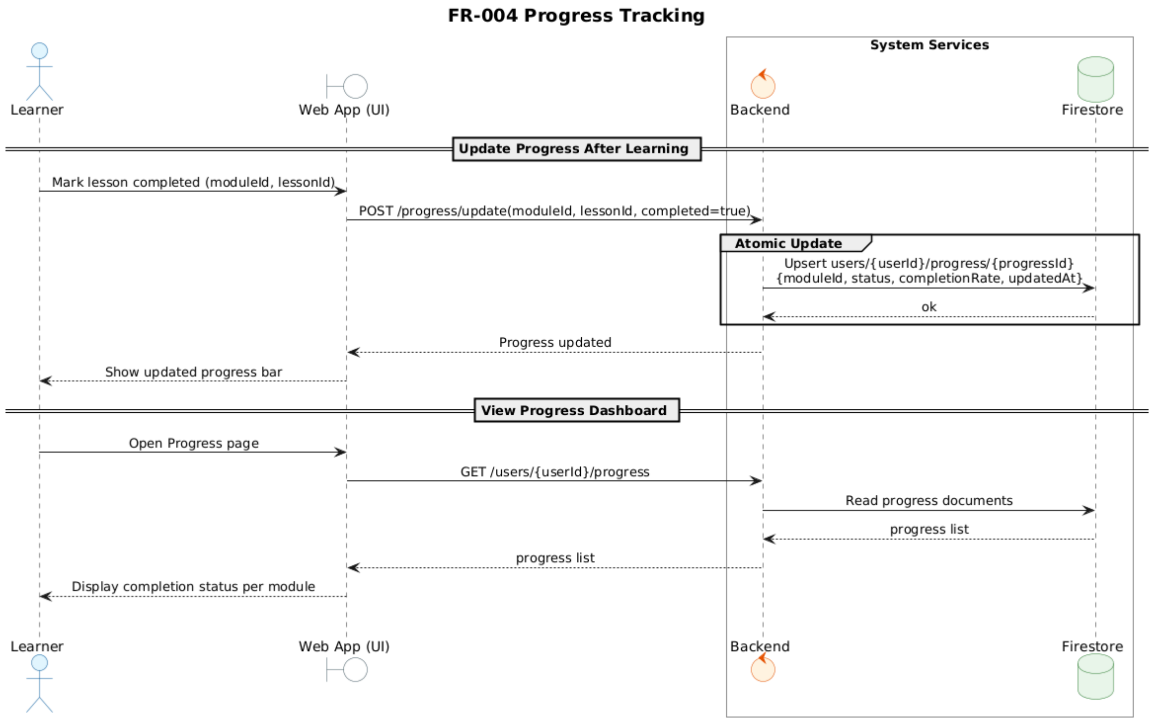

Figure 4.10: UML Sequence Diagram (FR-004)

4.6.5 UML Sequence Diagram (FR-005)

Figure 4.11 presents the UML sequence diagram for FR-005.

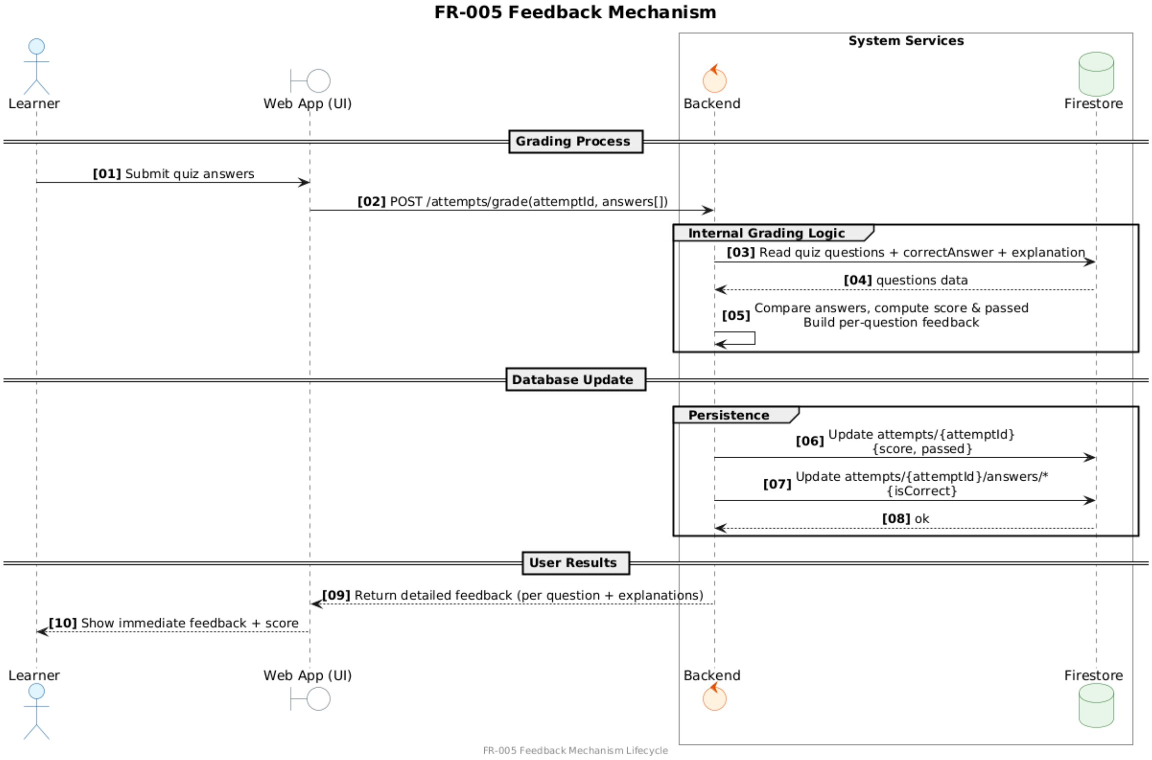

Figure 4.11: UML Sequence Diagram (FR-005)

4.6.6 UML Sequence Diagram (FR-006)

Figure 4.12 presents the UML sequence diagram for FR-006.

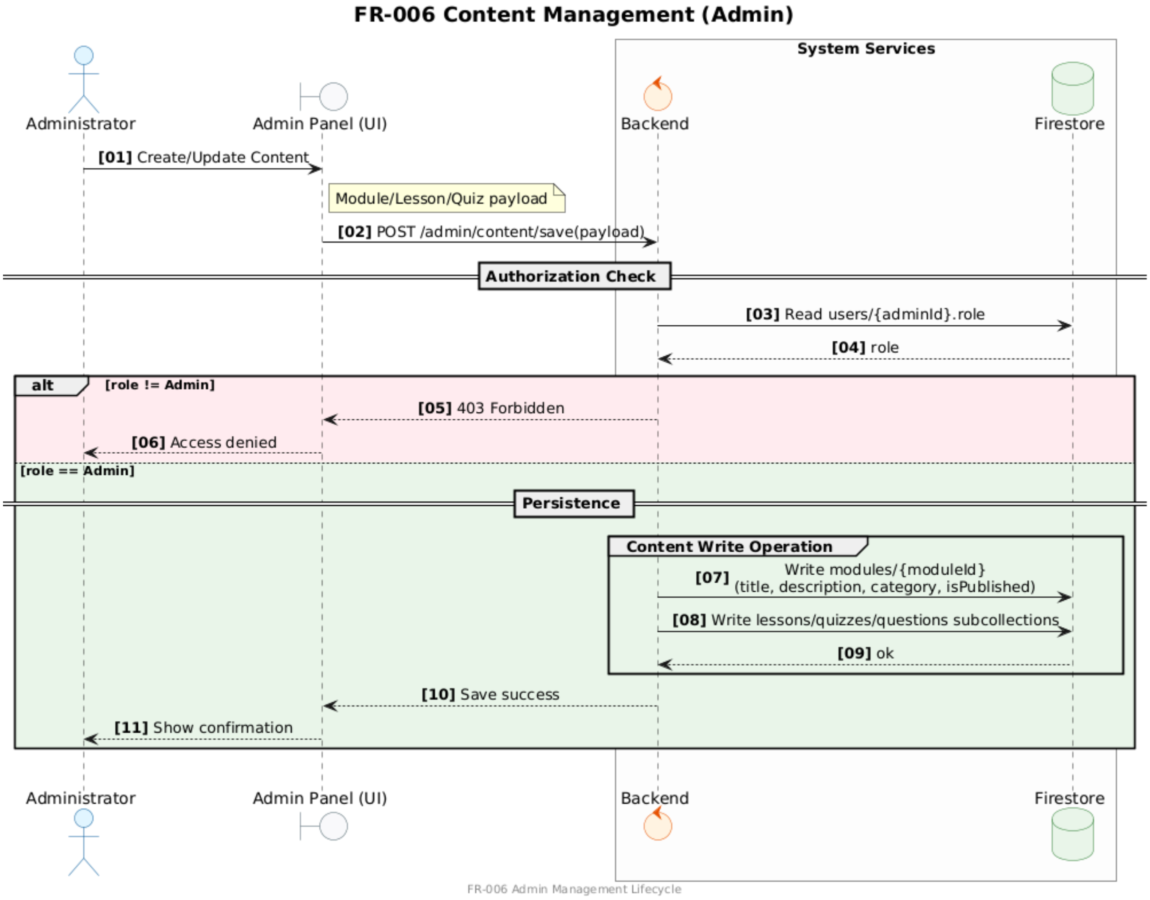

Figure 4.12: UML Sequence Diagram (FR-006)

4.6.7 UML Sequence Diagram (FR-007)

Figure 4.13 presents the UML sequence diagram for FR-007.

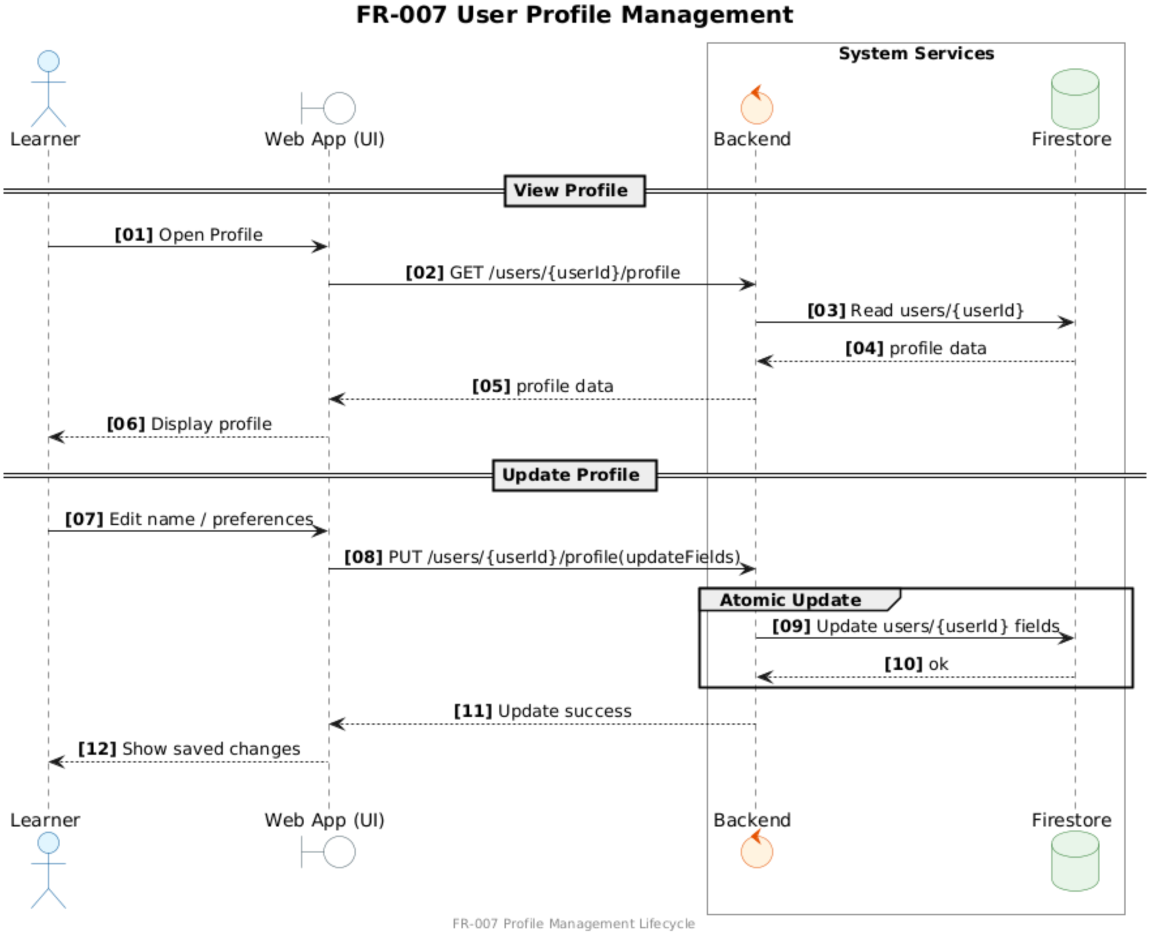

Figure 4.13: UML Sequence Diagram (FR-007)

4.6.8 UML Sequence Diagram (FR-008)

Figure 4.14 presents the UML sequence diagram for FR-008.

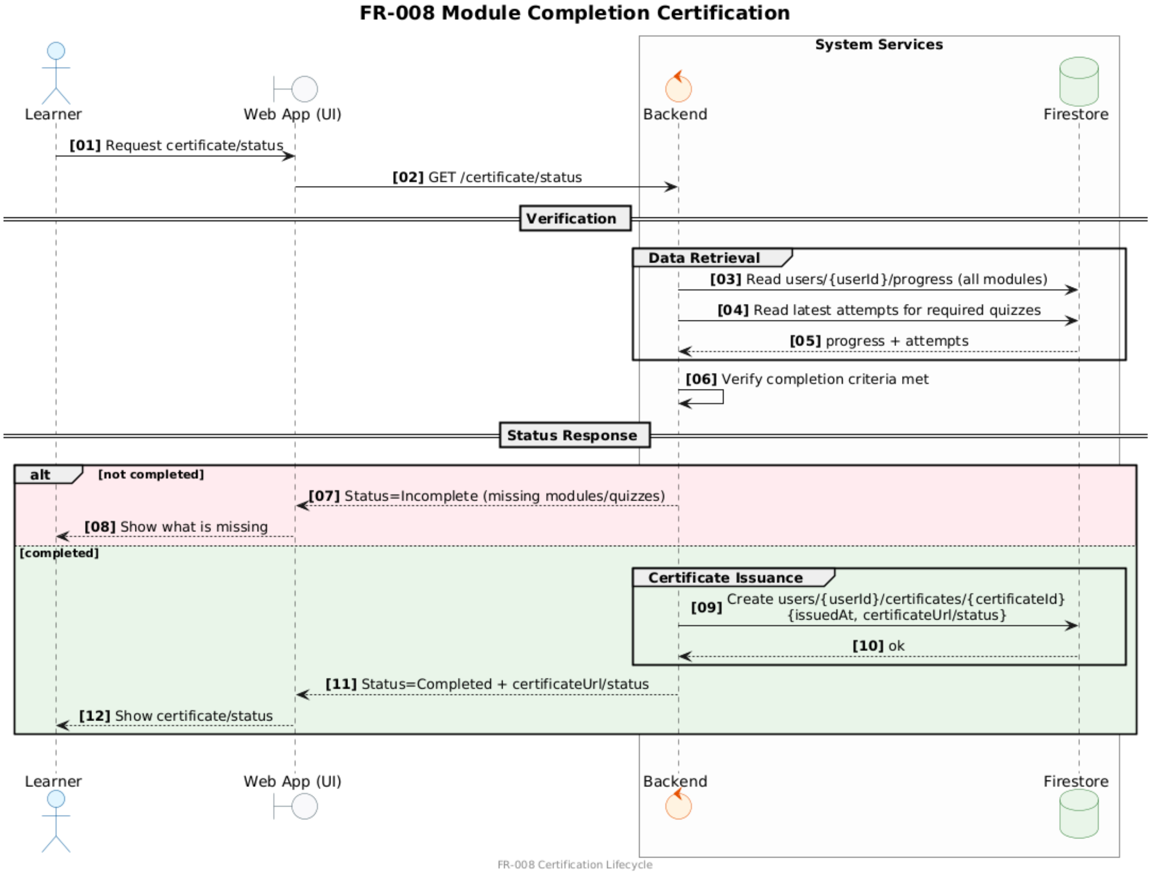

Figure 4.14: UML Sequence Diagram (FR-008)

4.6.9 UML Sequence Diagram (FR-009)

Figure 4.15 presents the UML sequence diagram for FR-009.

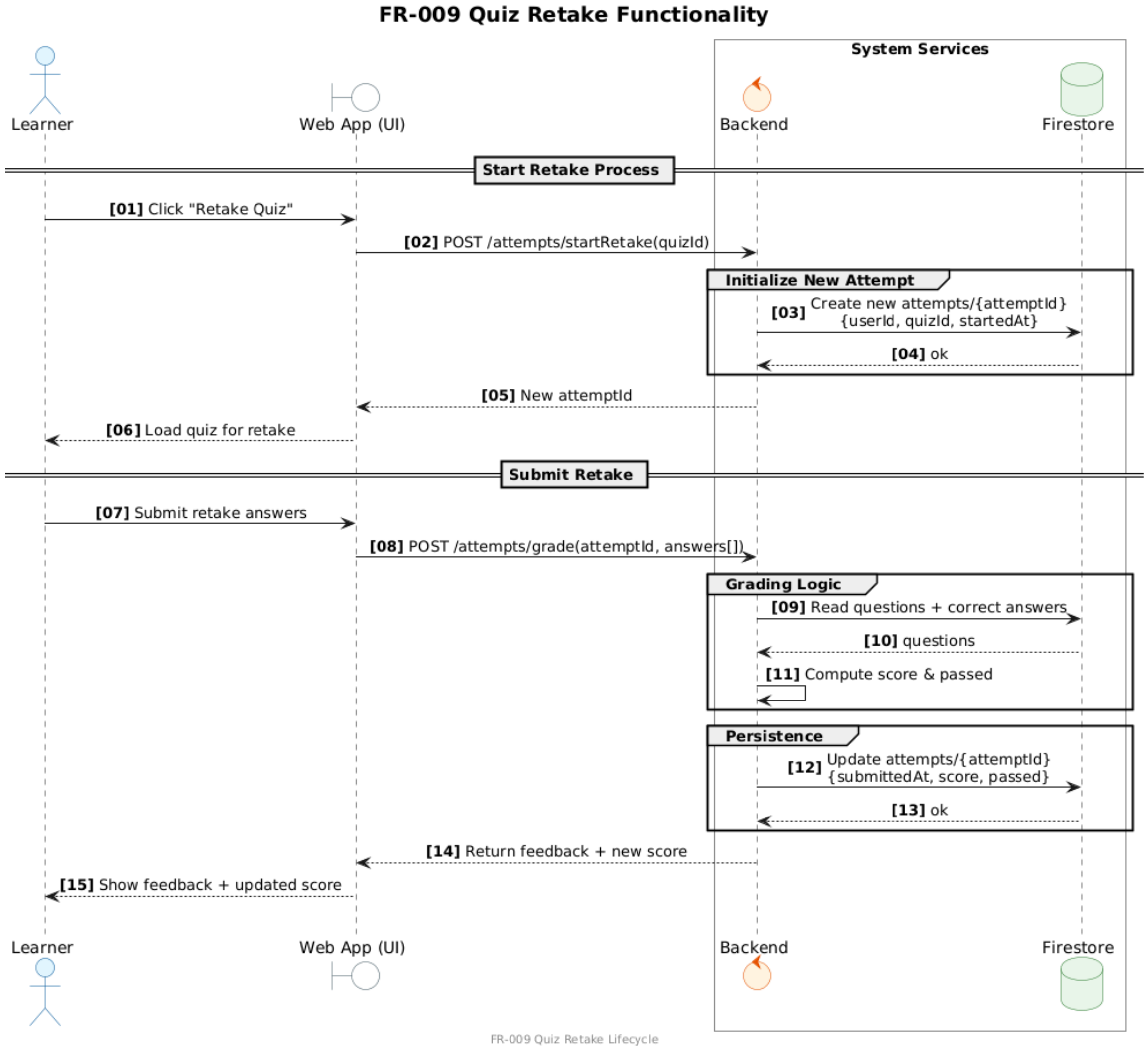

Figure 4.15: UML Sequence Diagram (FR-009)

4.6.10 UML Sequence Diagram (FR-010)

Figure 4.16 presents the UML sequence diagram for FR-010.

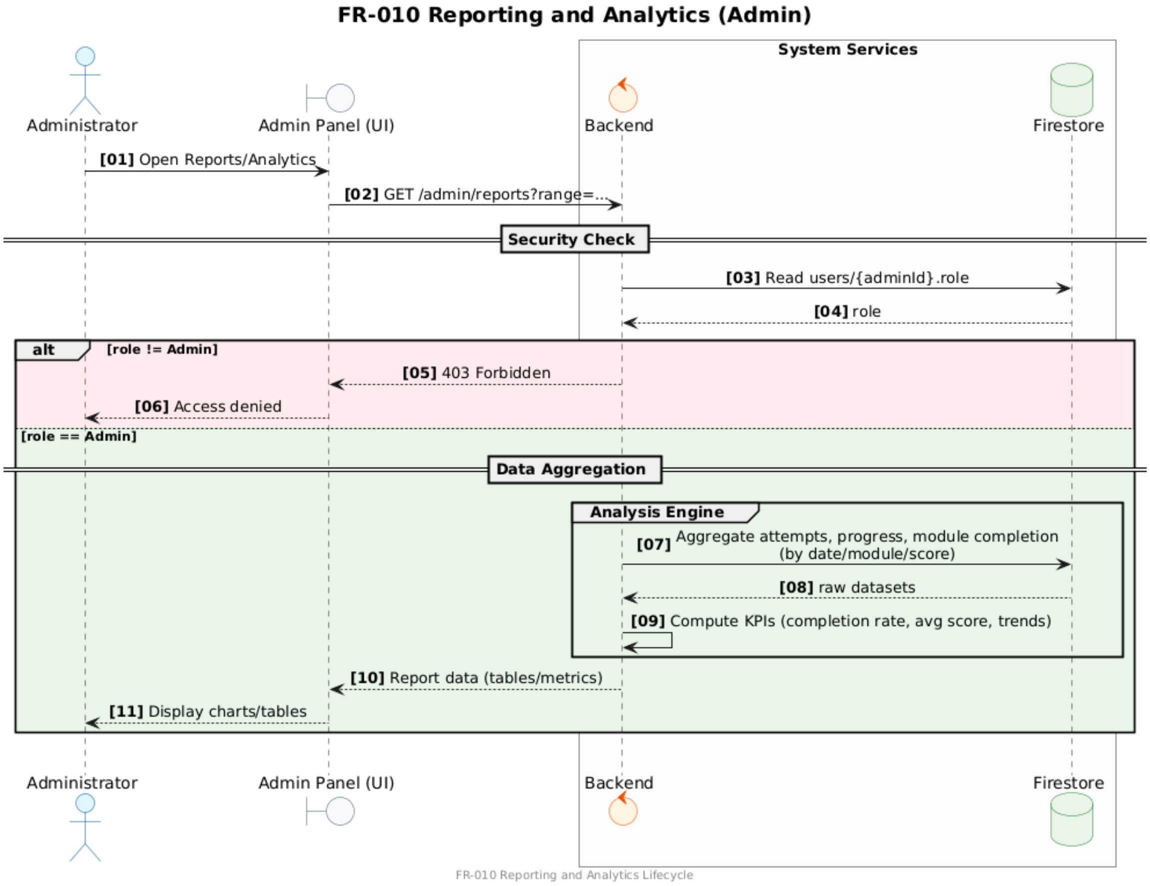

Figure 4.16: UML Sequence Diagram (FR-010)

4.6.11 UML Sequence Diagram (FR-011)

Figure 4.17 presents the UML sequence diagram for FR-011.

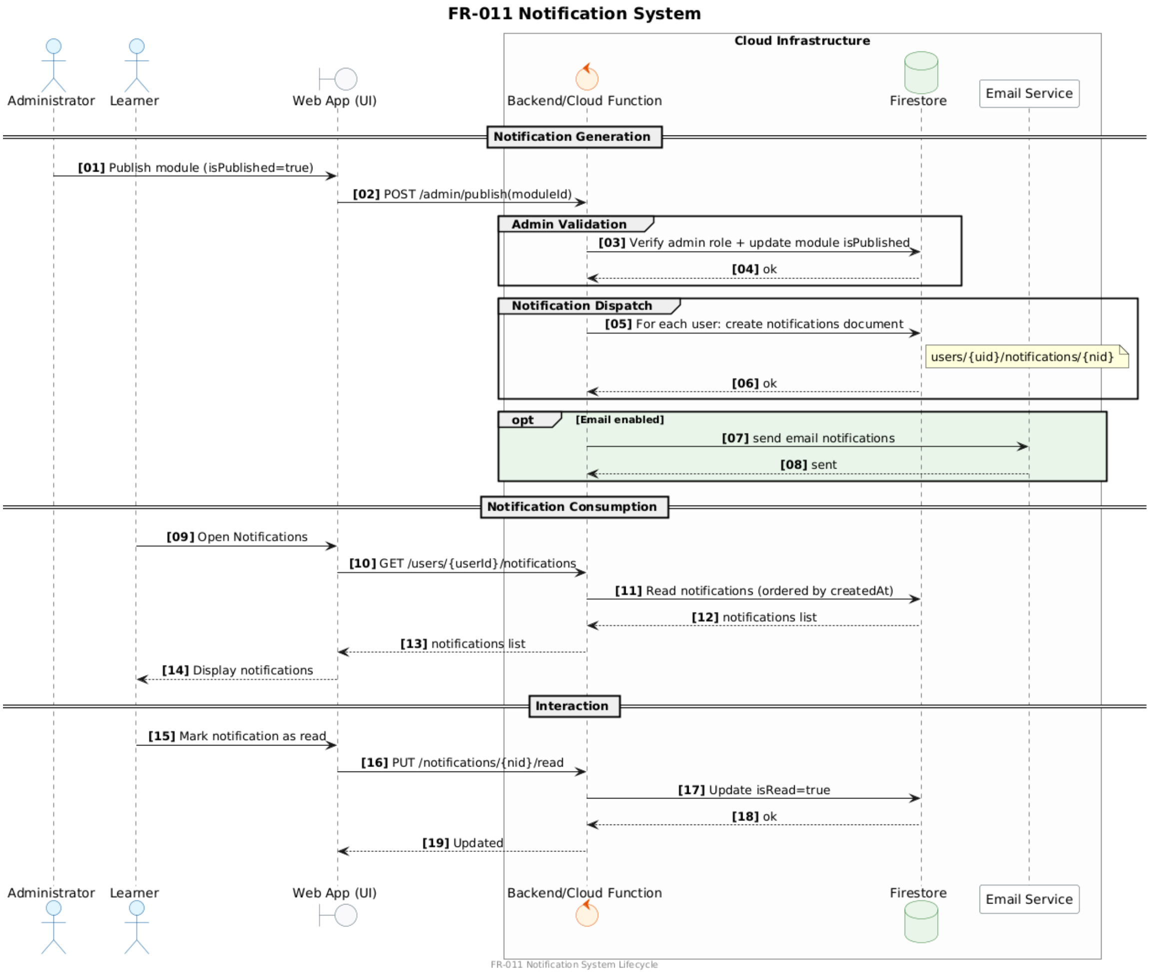

Figure 4.17: UML Sequence Diagram (FR-011)

4.6.12 UML Sequence Diagram (FR-012)

Figure 4.18 presents the UML sequence diagram for FR-012.

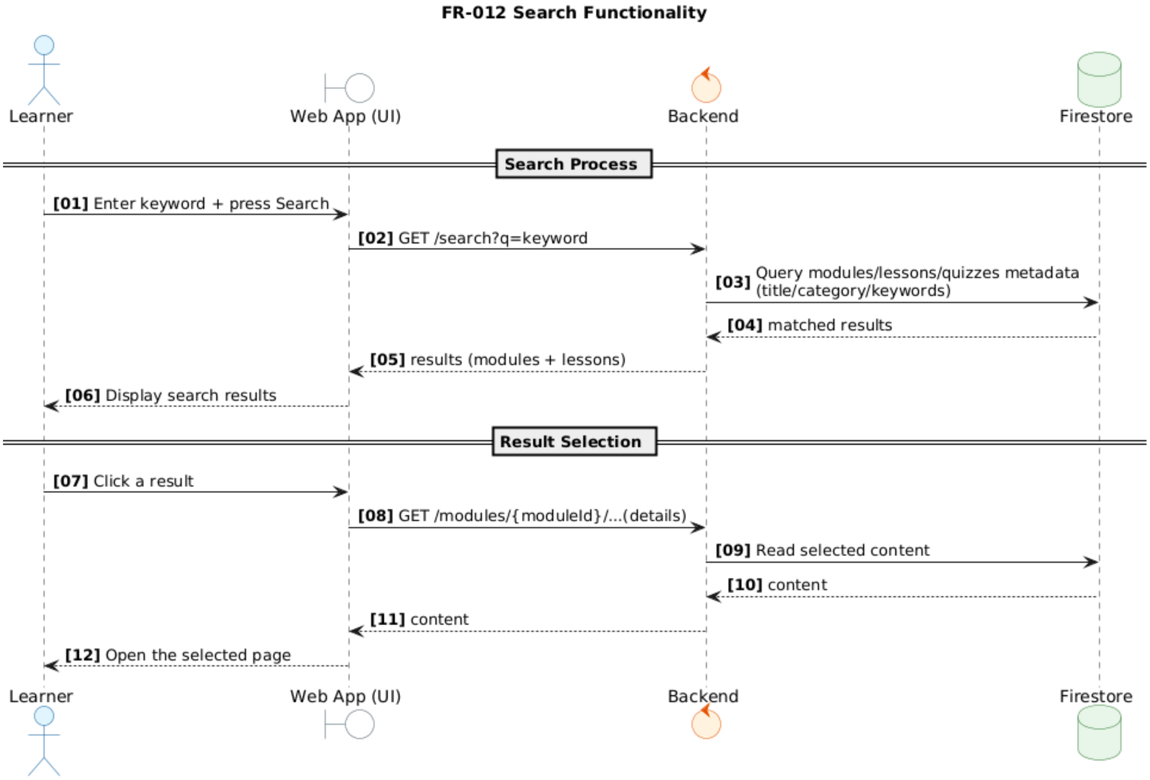

Figure 4.18: UML Sequence Diagram (FR-012)

4.6.13 UML Sequence Diagram (FR-013)

Figure 4.19 presents the UML sequence diagram for FR-013.

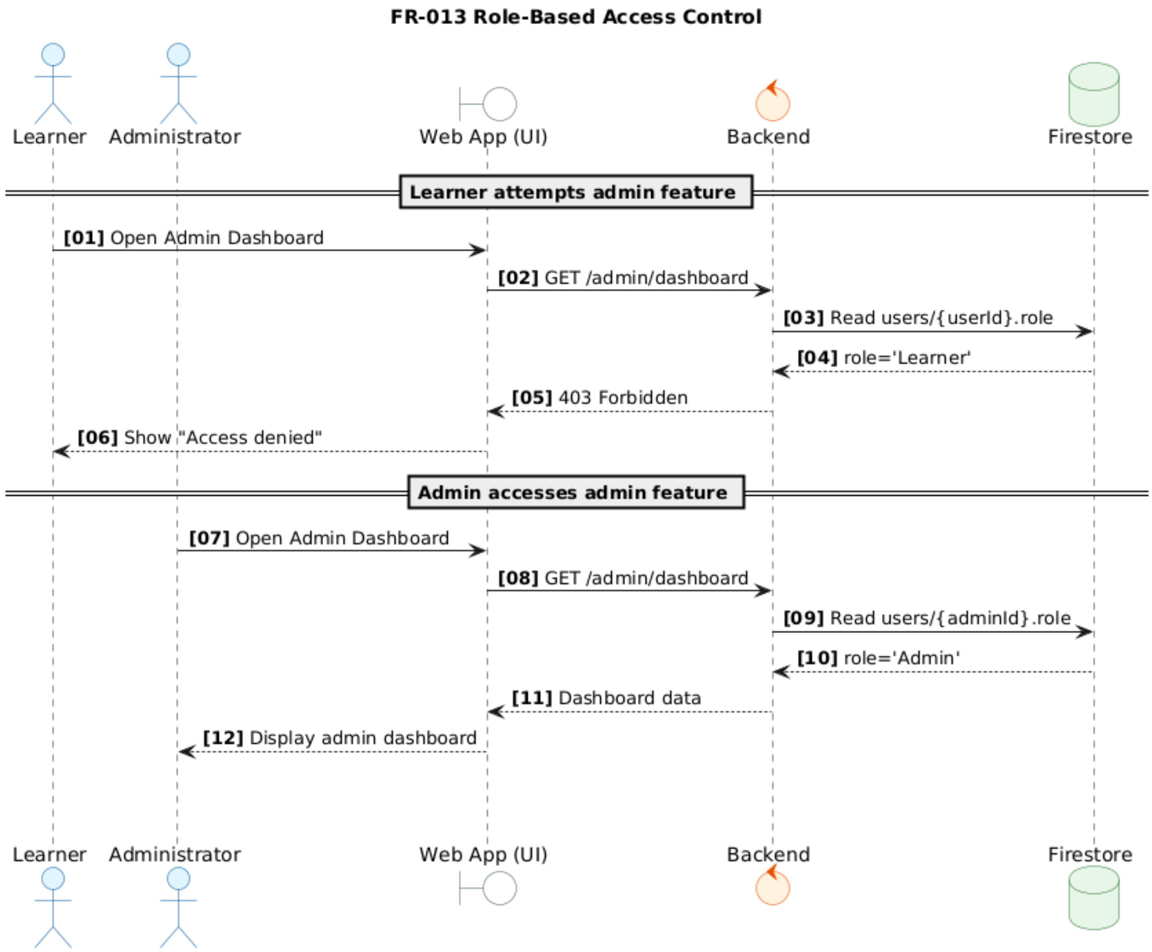

Figure 4.19: UML Sequence Diagram (FR-013)

4.7 UML Class Diagram

Figure 4.20 illustrates the UML class diagram, presenting the system's main classes, attributes, and relationships.

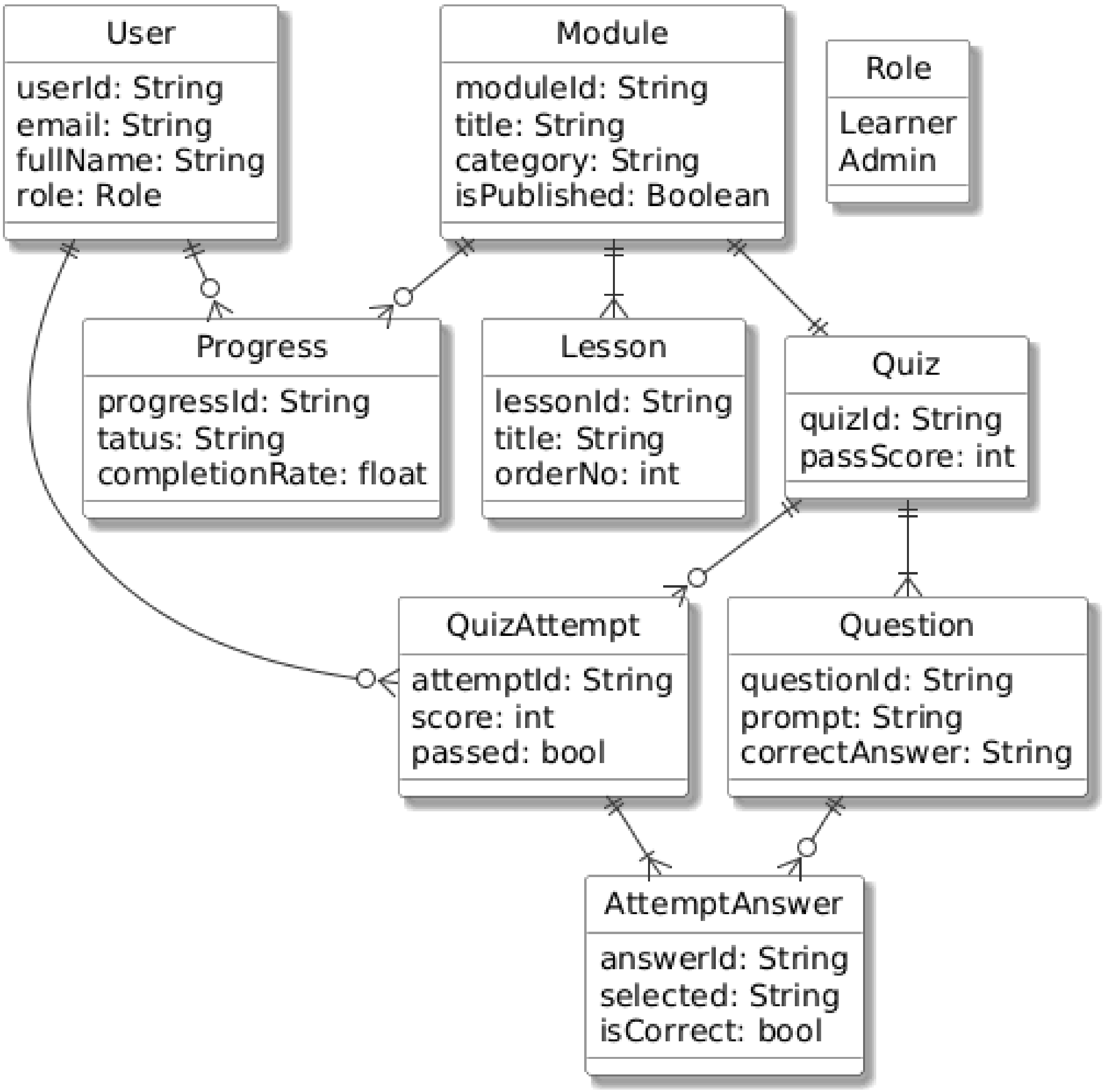

Figure 4.20: UML Class Diagram

4.8 Database Design

This section presents the database schema as separate blocks for clarity. Although MindShield uses Firebase/Firestore (collections and documents), the schema is presented in a table-like style to clearly show fields and relationships.

4.8.1 Database Design (Users)

Figure 4.21 presents the database design for the *Users* collection/table.

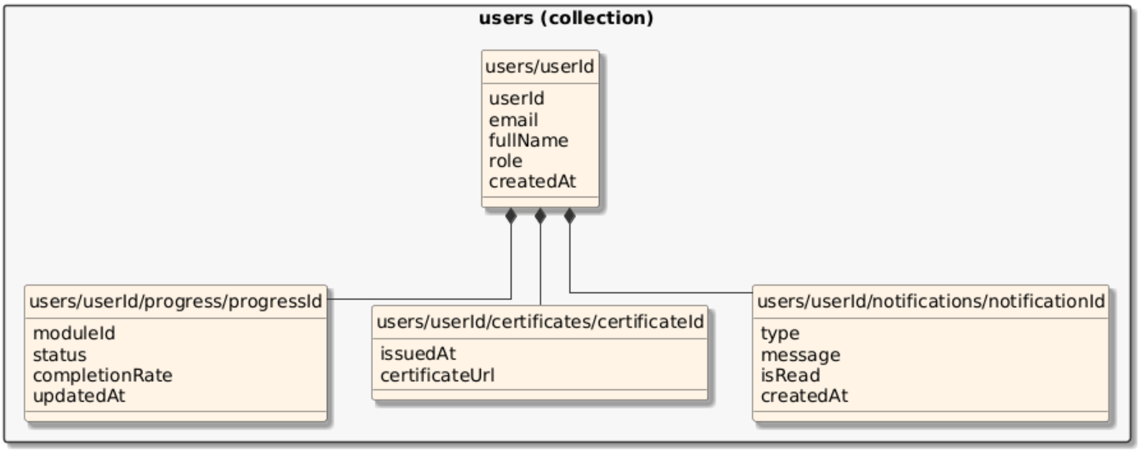

Figure 4.21: Database Design (Users)

4.8.2 Database Design (Modules)

Figure 4.22 presents the database design for the *Modules* collection/table.

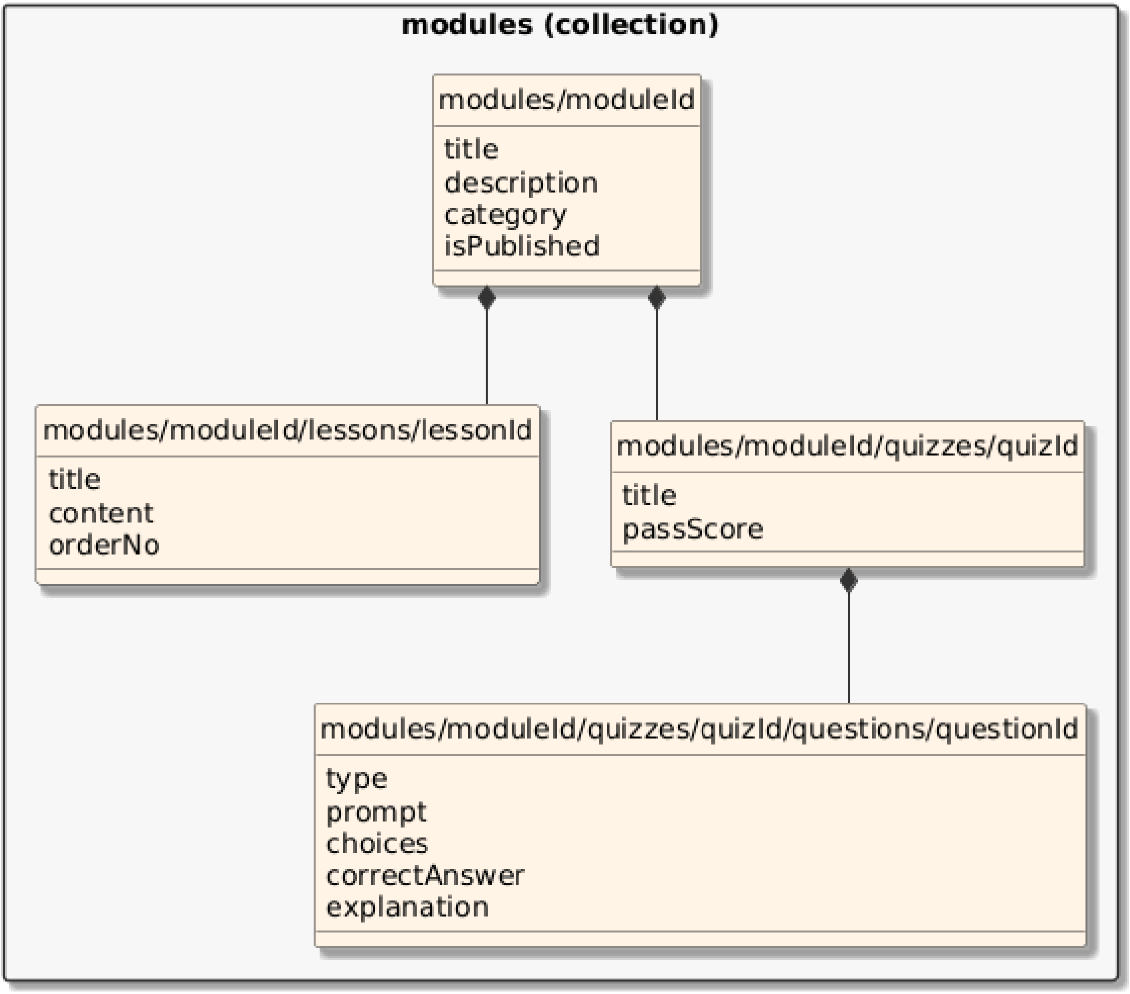

Figure 4.22: Database Design (Modules)

4.8.3 Database Design (Attempts)

Figure 4.23 presents the database design for the *Attempts* collection/table.

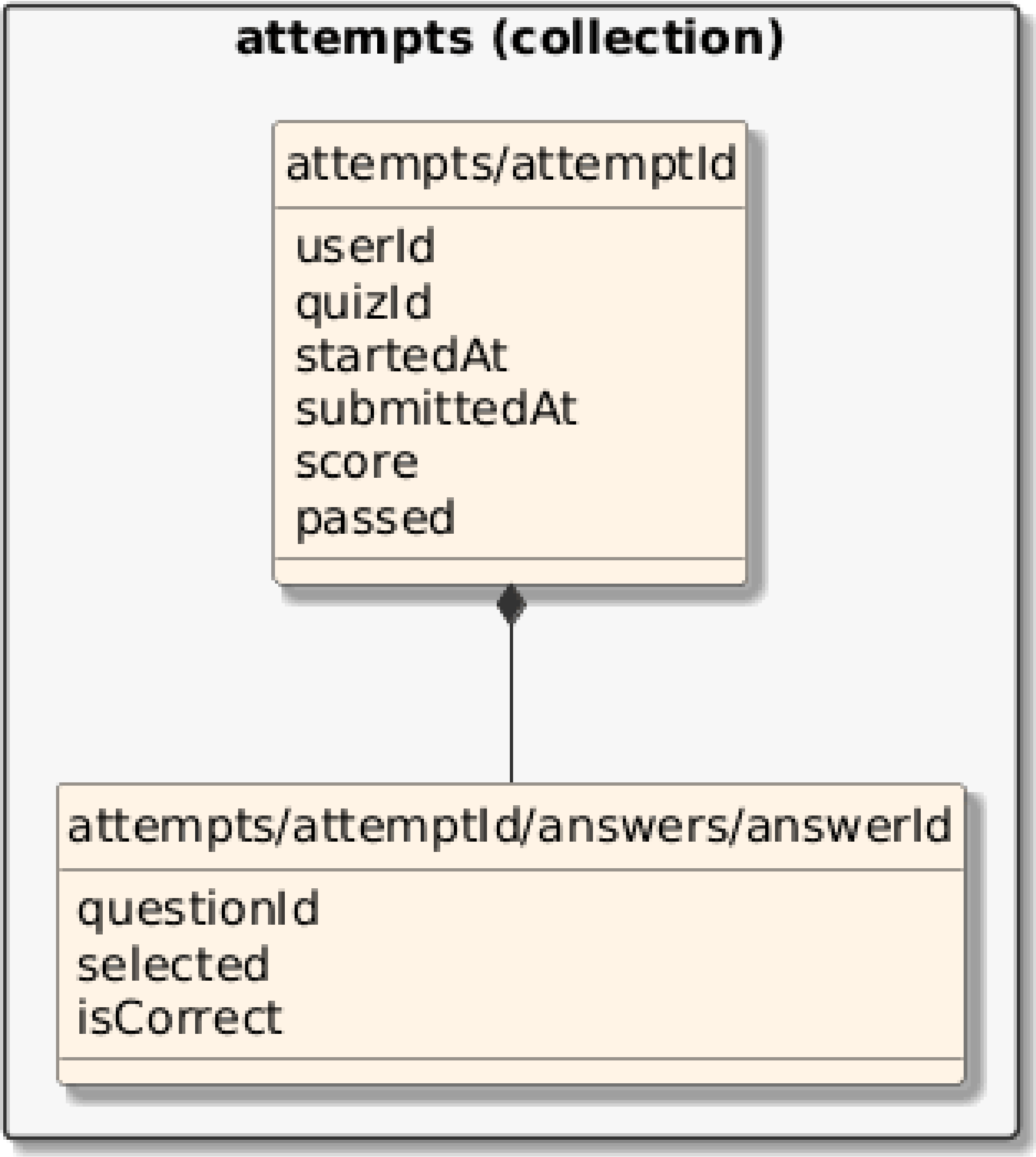

Figure 4.23: Database Design (Attempts)

4.9 Graphical User Interface Design (Low-Fidelity Prototype)

This section presents selected screenshots from the MindShield user interface to illustrate the overall look-and-feel and the main user journeys. The figures include the public landing page, the authentication page, and the administrator dashboard.

4.9.1 Public Landing Page

Figure 4.24 presents the main landing page of MindShield, highlighting the platform's value proposition and entry points for users.

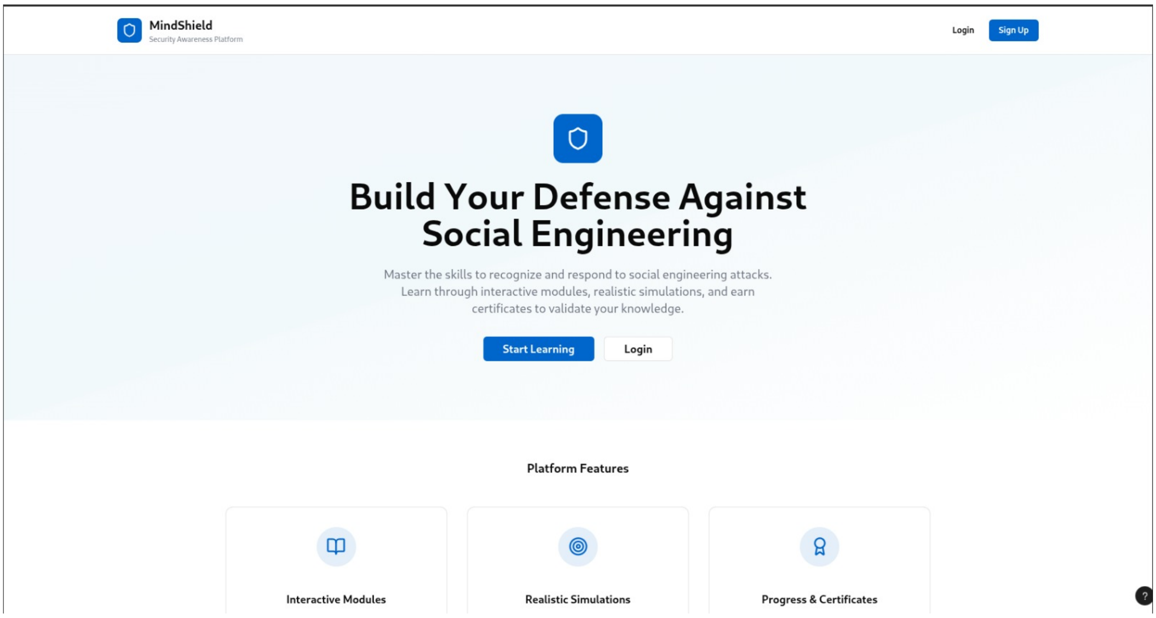

Figure 4.24: MindShield Landing Page (Main View)

Figure 4.25 shows the continuation of the landing page, emphasizing key learning topics and a call-to-action for account creation.

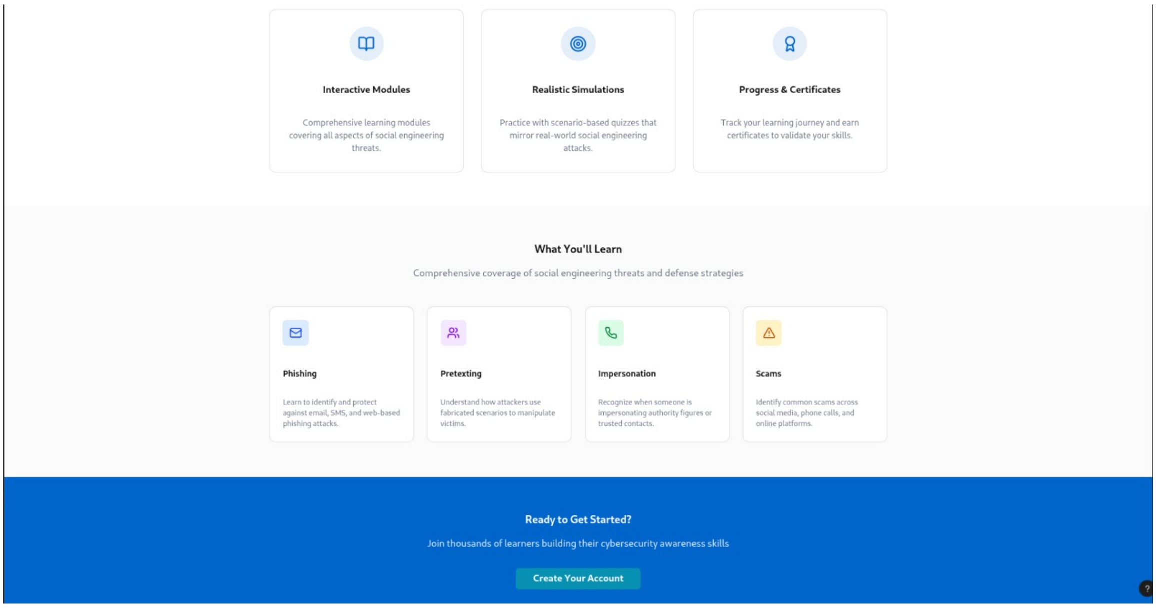

Figure 4.25: MindShield Landing Page (Features and Call-to-Action)

4.9.2 Authentication Page

Figure 4.26 presents the login interface used by both learners and administrators to access the system.

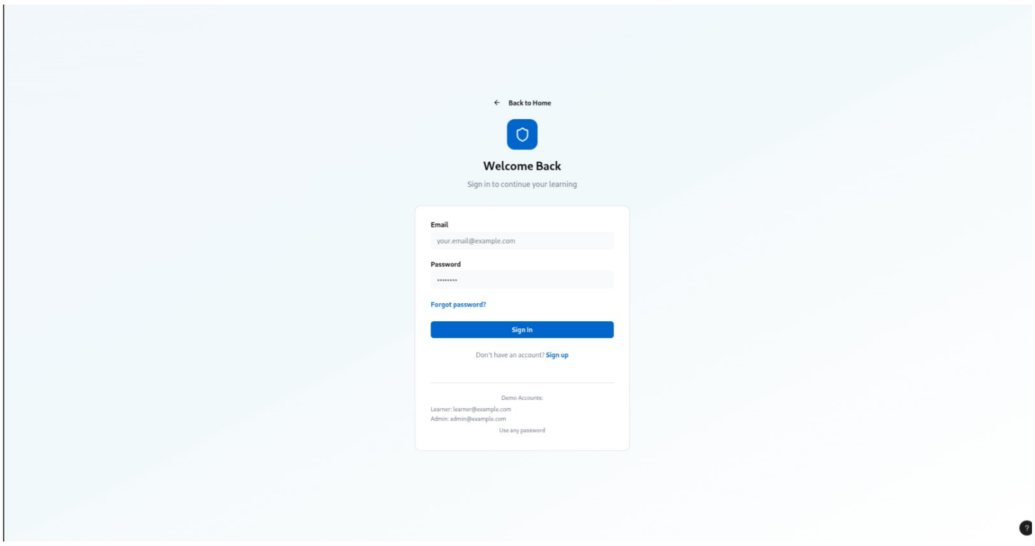

Figure 4.26: MindShield Login Page

4.9.3 Administrator Dashboard

Figure 4.27 presents the administrator dashboard, which provides key performance indicators and management navigation (content, users, and analytics).

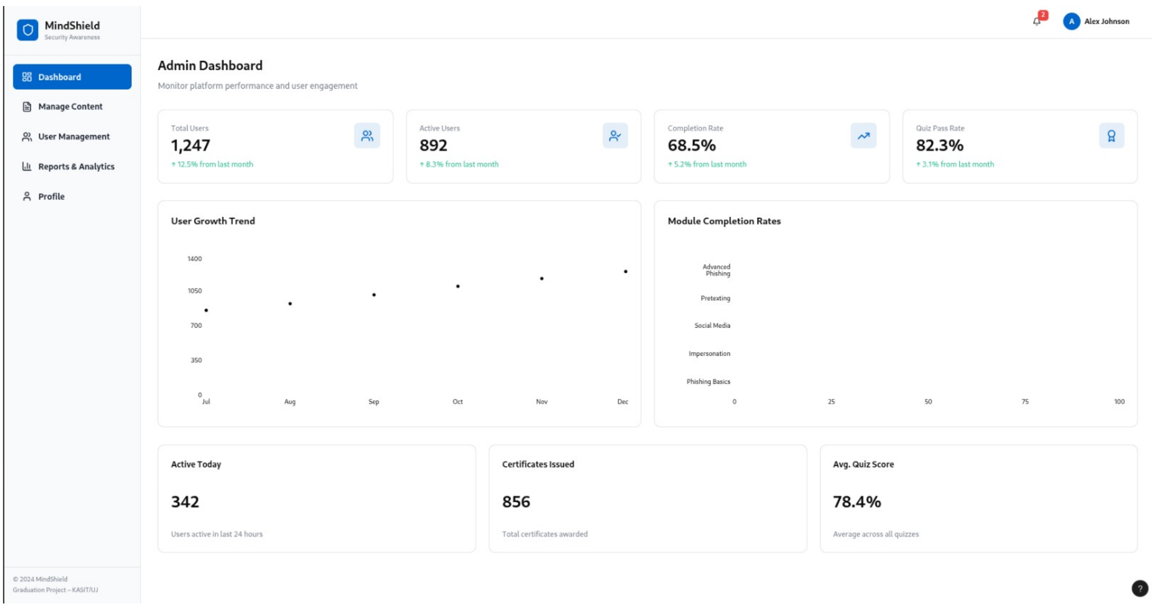

Figure 4.27: Administrator Dashboard Overview